Examination Bioinformatics

Linköpings Universitet, IDA, Statistik

Course:	732A51 Bioinformatics
Date:	2026/04/27, 8–12
Teacher:	Krzysztof Bartoszek
Provided aids:	The help material is included in the zip file **exam_help_material_732A51.zip**.
Grades:	A= $[18-20]$ points B= $[16-18)$ points C= $[14-16)$ points D= $[12-14)$ points E= $[10-12)$ points F= $[0-10)$ points
Instructions:	Provide a detailed report that includes plots, conclusions and interpretations. Give motivated answers to the questions. If an answer is not motivated, the points are reduced. Provide all necessary codes in an appendix. If you are asked to do plots, then make sure that they are informative, have correctly labelled axes, informative axes limits and are correctly described. Points may be deducted for poorly done graphs. Name your digital part solution files as: **[your exam account id]_[own file description]** If you have problems with creating a pdf you may submit your solutions in text files with unambiguous references to graphics and code that are saved in separate files There are **FOUR** assignments (with sub–questions) to solve. Include all code that was used to obtain your answers in your solution files. Make sure it is clear which code section corresponds to which question. Your code should be complete and readable, possible to run by copying directly into a script. Comment directly in the code whenever something needs to be explained or discussed. If you also need to provide some hand–written derivations please number each page according to the pattern: Question number . page in question number i.e. Q1.1, Q1.2, Q1.3,. . ., Q2.1, Q2.2, . . ., Q3.1, ...

Problem 1 (7p)

Define an HMM with the following parameters:
Three hidden states, S_1, S_2, S_3,
alphabet $A = \{1, 2\}$,

transition matrix for hidden layer $P = \begin{bmatrix} 1 & 0 \\ 1/4 & 3/4 \end{bmatrix}$, initial distribution $\pi = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$,

emission probabilities by hidden states ($b_i(j)$ is the probability that a hidden state that equals i will emit observed state j):
$b_1(1) = 0$, $b_1(2) = 1$,
$b_2(1) = 1/4$, $b_2(2) = 3/4$.

What are all possible state sequences for the following observed sequences $\mathcal{O}$, and what is the probability of each observed sequence?
(a) $\mathcal{O} = 1, 2, 1$
(b) $\mathcal{O} = 2, 1, 1$
(c) $\mathcal{O} = 2, 2, 2$

Problem 2 (4p)

In the file `AJ250102_1.fasta` you will find part of the DNA deposed in GenBank entry AJ250102.1. The provided DNA is a partial (e.g., final STOP codon can be missing) sequence for rabbit's haptoglobin protein. The goal of this exercise is to find the protein code, i.e., correctly translate the nucleotide sequence into the protein sequence. Your final provided sequence (in a fasta file) should be only the sequence of the protein without any extras. Explain what you did, why and motivate why you think your translation is correct. TIP: You might want to look at **ape**'s functions: `ape::complement()`, `ape::read.FASTA()`, `ape::trans()` and the options they have.
The correctly translated protein sequence should contain the substring GAVITLLL. Take note that you may **NOT** use the presence of this substring to motivate correctness of your translation.

Problem 3 (5p)

In Fig. 1 (next page) you have a phylogeny provided. Encode this tree *manually* in the `R` **ape** package's `phylo` format. Afterwords write code that plots your `phylo` object exactly (with the exception that the branch lengths need not be written on the plot) as it is in Fig. 1 (you will need to read the documentation of `?plot.phylo`).

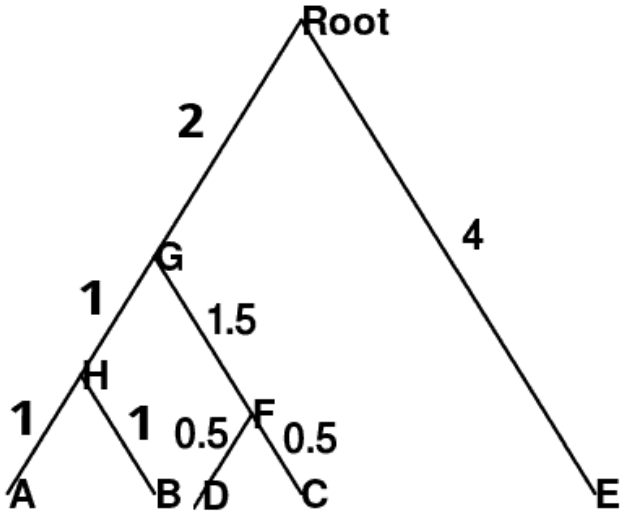


Figure 1: Phylogeny for Problem 3. The numbers next to the branches are the branch lengths. Next to each node and tip, the label of the node/tip is provided.

Problem 4 (4p)

In Fig. 2 you have a graph describing a particular food network. An arrow from species (node) x to y means that species x is eaten by species y. Calculate the clustering coefficient of the graph and the betweenness for each node. Is this food network robust to the loss of some species? If not, indicate which ones are critical. What do the previously calculated clustering coefficient and the betweenness values tell us about the robustness of the network?

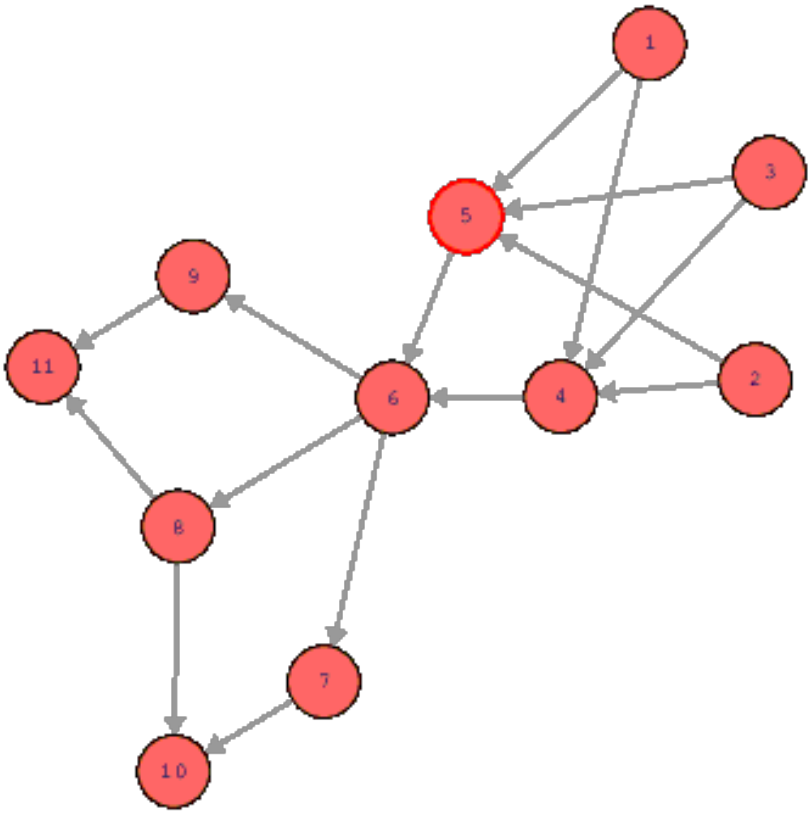


Figure 2: Food network for Problem 4.